Marmara University, Faculty of Engineering
Department of Electrical and Electronics Engineering

Mühendislik Fakültesi

EE4053 –Microwaves, Spring 2025

Students will develop a solid understanding of the components encountered in typical high-frequency systems and develop skill in the analysis and design of microwave circuits incorporating transmission lines.

The key topics of this course are,

1. Introduction to RF, introduction to the dimension and units, frequency spectrum, power calculation, free space propagation.
2. AWR basics, and simulation of circuits
3. Analysis of Transmission Lines. Characteristic impedance definition.
4. Microstrip transmission lines and their termination.
5. Introduction to the Smith Chart. AWR Smith Chart Experiment for parallel and series connections of lumped elements.
6. Introduction to the Impedance transformation
7. Scattering Parameters, Measurement Basics and instruments (Vector Network Analyzer, Spectrum Analyzer, etc.). VNA Calibration.
8. Matching Networks. Practical examples on matching Networks
9. RF Amplifier Basics: Power Gain, Stability, Noise
10. RF Amplifier Basics: Stability and Gain Circles
11. RF Amplifier Design- theory
12. RF Amplifier Design- simulation
13. RF Amplifier Design- characterization
14. An overview of RF Filter Design

Instructor: Prof. Dr. Serkan Topaloğlu, serkan.topaloglu@marmara.edu.tr

Course Hours:

Lecture: Wednesday, 15:00 – 16:20 /20 mins. Break / 16.40-18.00 (RTEM4.Z09)

Textbooks:

- RF Circuit Design: Theory and Applications: by R. Ludwig, G. Bogdanov, 2nd Edition, Pearson Education, 2009, ISBN: 978-0-13-135505-7
- Gonzalez, Guillermo, Microwave Transistor Amplifiers: Analysis and Design, 2nd edition. Prentice-Hall, 1997
- S.A. Maas, Nonlinear Microwave and RF Circuits, 2nd Edition, Artech House, Inc., 2003. Microwave Engineering, 3rd ed., David M. Pozar, JohnWiley and Sons, 2005

Grading:

There will be assignments and projects for the midterm exam and 1 final exam. There will be projects and assignments; and the cumulative grades of these works will be used as Midterm Exam score.

Midterm (Projects&Assignments): 50 % (min.5 project and assignments)*

**must to submit each assignments and projects to be able take the final exam.*

Final Exam: 50 %**

***Attendance is mandatory, and students who do not meet the attendance requirement will not be allowed to take the final exam.*

Overall score will be converted to a letter grade as given in the below table:

AA: 85 -100
BA: 75 - 84
BB: 65 - 74
CB: 55 - 64
CC: 50 - 54
DC: 40 - 49
DD: 35 - 39
FD: 30 - 34
FF: <30

Additional Remarks:

Students should register AWR university program and get a free educational AWR license for 6 months. Most of the homework and course projects will contain designing and simulation of various RF circuits.

Last Update: 20.02.2025